

A STUDY ON IMPROVING MULTI-CLASS AUDIO SOURCE SEPARATION VIA DECOUPLED CLAP QUERY OPTIMIZATION AND AN AUTOMATED DATA ENGINE

Amirhossein Hajavi, Hanhee Lee^{}, Pushya Jain^{*}, Sky Qiao, Emmanuel Ko, Yuanhao Yu, Irina Kezele*

Noah’s Ark Lab, Huawei Technologies Canada co.
{amirhossein.hajavi, irina.kezele}@huawei.com

ABSTRACT

Language-queried audio source separation (LASS) enables extracting any sound source using natural language. However, adapting LASS models to application-specific sound classes is challenging due to noisy training data and limited semantic coverage of the CLAP-based control signals. We propose a framework comprised of an automated data engine for training-data curation and a two-stage optimization process for class-specific CLAP control signals. Our objective evaluations across seven sound classes show that data refinement and control signal optimization consistently improve source separation performance. Subjective evaluation with 17 participants further demonstrates perceptual improvements of the model trained with optimized control signals over baseline and similar commercial models.

Index Terms— Sound source separation, Queried Sound Source Separation, CLAP, Automated Data pipeline

1. INTRODUCTION

Universal audio source separation aims to isolate a mixture of multiple arbitrary sound sources to its components [1]. Among its various paradigms, language-queried audio source separation (LASS) uses natural language as the control signal to extract a target sound source. The emergence of Contrastive Language-Audio Pre-training (CLAP) [2] has led to considerable advancements in the performance of LASS models [3, 4, 5, 6, 7, 8, 9, 10, 11]. By providing semantically aligned language and audio embeddings, CLAP enables effective language-based control signals, resulting in consistent improvements in source separation performance across both objective and subjective evaluations.

Audio source separation models are often deployed in application domains where a subset of sound classes is of primary interest. As a consequence, improving the separation quality of these specific classes becomes particularly important. Historically, application-specific source separation has primarily focused on tasks such as music source separation [12, 13, 14, 15] or target speech separation [16]. The flexibility afforded by text encoders such as CLAP enables arbitrary

sound classes to be specified using natural language rather than predefined label sets. Consequently, pretrained LASS models can be adapted to a much broader range of downstream applications [17].

Identifying suitable target sound classes for downstream applications remains challenging. Although LASS models perform inference using natural language prompts, their performance on a target class depends heavily on how well that class is represented in the training data. In some cases where a general set of sound classes are selected as target sound classes, the collective samples of audio that are subcategories of the general classes can be used for training the source separation model. However, embeddings learned solely through CLAP do not necessarily preserve the hierarchical relationships between sound classes therefore can not be used effectively as control signals.

Existing work primarily improves source separation performance through architectural modifications to model training and inference [17, 18]. Meanwhile, the challenges of curating useful training data and optimizing language-based control signals remain underexplored. Motivated by these limitations, we make the following contributions: (1) We propose a multi-stage automated data engine for collecting useful training data from the available large-scale datasets and validating their relevance to the target classes. (2) We propose an enhanced control signal by aggregating multiple CLAP embeddings for each class and decoupling the optimization of the resulting embeddings from the end-to-end pipeline to increase inter-class discriminability for general sound source categories. (3) Through extensive experiments on the 7 sound classes, we show that both our data engine and control-signal optimization significantly improve the performance of the source separation models for the target classes.

2. RELATED WORK

Early works on sound source separation focused on music source separation [12, 13, 14, 15] or target speech separation [16]. The benchmarks defined for these tasks either included a predefined set of sound classes (e.g. four musical instrument classes in [19]) or a reference speech signal from the target speaker [20]. Universal audio source separation, on the

^{*} Equal contribution

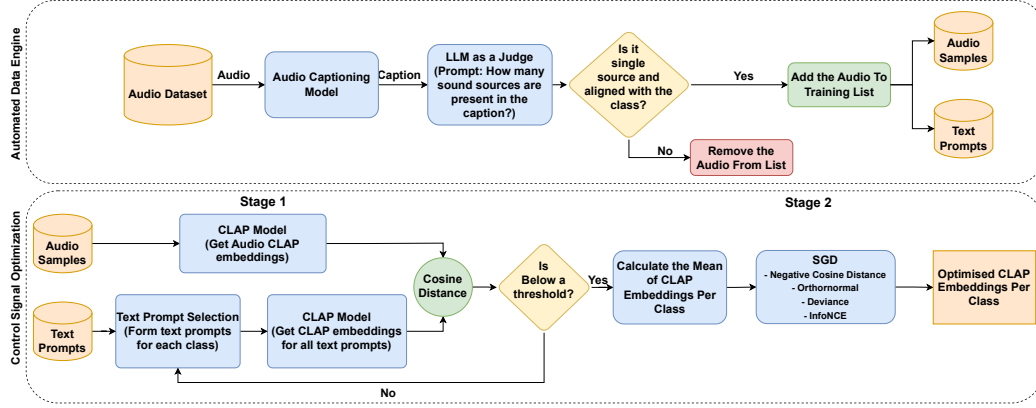


Fig. 1. An overview of the proposed framework. (Top) The pipeline for the automated data engine which consists of an audio captioning and LLM-as-a-Judge component. (Bottom) The optimization process for control signals using an iterative procedure to select the candidate CLAP embeddings followed by a SGD optimizer.

other hand, removes this restriction by allowing arbitrary target sound classes[1]. The initial models that aimed to perform this task estimated multiple masks for the input mixture to extract the different sound sources individually [1]. More recently, studies such as [21, 22] proposed models that perform single-target source separation using natural language as the control signal. These models adopted an encoder-decoder architecture conditioned through *Feature-wise Linear Modulation* (FiLM) [23]. This enabled the source separation models to isolate any arbitrary sound class given an appropriate control signal. While the early LASS models relied on generic sentence encoders, the adoption of CLAP in later work provided semantically aligned language-audio embeddings as the control signal that substantially improved source separation performance [3, 4, 5, 6, 7, 8, 9, 10, 11].

A limited number of studies have investigated improving LASS performance for specific subsets of sound classes. The work done in [17] focuses on improving the separated signals of specific target classes after inference, using a generative adversarial network (GAN) to refine the output of the source separation model. Another line of work combines domain partitioning and parameter efficient finetuning (PEFT) to adapt LASS models to multiple subsets of sound classes and subsequently combine the adapted models into a single model [18]. Although these studies consider specific subsets of the sound classes, they do not specifically focus on refining the training data or optimizing the control signal. In this work, we aim to improve LASS performance for a set of defined sound classes through introducing a multi-stage automated data engine for training data curation and an optimized set of control signals for better separation.

3. METHOD

We propose a framework for adapting LASS models to subsets of sound classes: a multi-stage data engine for constructing and filtering training data, and a control-signal optimiza-

tion process for refining class-specific CLAP embeddings. Together, these components are used to fine-tune a pretrained LASS model to improve separation performance for target sound classes. An overview of the proposed framework is shown in Fig 1.

3.1. Data Generation and Filtering Pipeline

Large-scale audio datasets often contain mislabeled samples or samples where multiple sound classes are present but only one sound class is specified by the label. During the training, the presence of other non-target audio classes in the ground-truth target signal can negatively affect the separation performance of the model. The proposed data engine first uses an audio captioning model which identifies different sound sources in audio samples and provides a detailed caption describing each sample. The caption is then given to an LLM-based judge which is responsible for checking two criteria: (a) Does the audio include only a single sound class? (b) Does the caption fall under any of the defined target sound classes?

The use of audio captioning models allows the data engine to leverage datasets that do not provide detailed descriptions of their audio samples. This allows us to draw training samples from datasets collected with a wide range of purposes rather than the only the datasets with detailed annotations. For this study, we collected samples from multiple datasets, including AudioSet [24], AudioCaps [25], VGG-Sound [26], Freesound [27], and BBC Sound Effects platform [28]. When existing captions are available, or when multiple captioning models are used, all the generated and pre-existing captions are combined creating a comprehensive description of each audio sample¹.

The comprehensive description is then incorporated into the body of the prompt provided to the LLM-based judge, together with the specified subset of target classes. The judge

¹The details and the prompts used in this experiment are published in <https://amhajavi.github.io/AudioSourceSeparation>

then evaluates whether: (1) The description indicates the presence of only a single sound class or mentions any other classes. (2) Whether that class belongs to the target set. Samples that do not adhere to these criteria are removed from the candidate pool ¹.

3.2. Control Signal Optimization

CLAP is trained using a contrastive objective to learn semantically aligned representations of audio and text. As a result, CLAP embeddings are commonly used as a control signal for LASS models, providing an effective means of specifying target sound classes through natural language. However, when adapting a LASS model to the defined sound classes of interest, CLAP embeddings may not adequately capture the semantic coverage of the subcategories within each target class. This can make it difficult for the model to associate certain classes as subcategories of a target class. The proposed control signal optimization provides a solution to this problem through a two-stage process that refines the class-specific CLAP embeddings to improve their ability to represent the target class and its subcategories while maintaining separability from other classes.

The first stage consists of an iterative selection process that selects the largest subset of text prompts subject to a constraint on the cosine distance of the mean text and audio CLAP embeddings. We start with a large set of possible textual descriptions for each sound class and a fixed set of audio samples best representing each sound class. Through this process, we maximize the number of retained prompts per target sound class while maintaining audio-text semantic alignment between the remaining text prompts and the set of audio samples. For a given target sound class, we define $D_T^c \in \mathbb{R}^{n \times d}$ and $D_A^c \in \mathbb{R}^{m \times d}$, where n is the number of textual descriptions designated for class c , m is a fixed number of audio samples chosen representing the sound class, and d is the size of the CLAP embedding vectors. The subset of the candidate textual prompts for each class S^c is selected through:

$$S^c = \arg \max_{S^c \subseteq \{1, \dots, n\}} |S^c| \quad \text{s.t.} \quad \sum_{i=1}^m \left(1 - \frac{\overline{D_T^c[S]} \cdot D_{A_i}^c}{\|D_T^c[S]\| \|D_{A_i}^c\|} \right) \leq t \quad (1)$$

where $\overline{D_T^c[S]}$ denotes the mean CLAP embeddings of the text prompts indexed by S and t is the threshold of the acceptable cosine distance between the mean text and audio embeddings and is selected by examining the trade-off between the number of retained text prompts and audio-text alignment. After selecting S^* , the retained text prompts are used to construct the class-specific control embedding in the second stage.

The second stage consists of an optimization procedure to construct the control signals from the selected subset of

CLAP embedding candidates. Each class is then represented by an embedding vector, e_c , which is the mean vector over all the embedding vectors of S^c . The embeddings are then optimized using a stochastic gradient descent (SGD) optimizer with respect to three objectives:

1. Orthogonal Normalization ($\mathcal{L}_{\text{Ortho}}$) which aims to maximize the angular distance of the embedding vectors through

$$\mathcal{L}_{\text{Ortho}} = \|e_c e_c^T - \mathbb{I}_{n \times n}\|^2 \quad (2)$$

where $\mathbb{I}_{n \times n}$ is the identity matrix.

2. Regularization through Deviation ($\mathcal{L}_{\text{Dev}}$) which aims to limit the deviation of the optimized embeddings from the original embeddings $\hat{e}_c$ through

$$\mathcal{L}_{\text{Dev}} = \|e_c - \hat{e}_c\| \quad (3)$$

3. Information Noise-Contrastive Estimation ($\mathcal{L}_{\text{INCE}}$) [29] which performs a contrastive optimization between the embedding and the embeddings belonging to different classes.

$$s(x, y) = \frac{x^T y}{\tau}$$

$$\mathcal{L}_{\text{INCE}}^c = \mathbb{E}_{e_c, D_A^+, \{D_{A_i}^-\}} \left[-\ln \frac{e^{s(e_c, D_A^+)}}{e^{s(e_c, D_A^+)} + \sum_{i=1}^M e^{s(e_c, D_{A_i}^-)}} \right] \quad (4)$$

where τ is the temperature value which is adopted from [29] and D_A^+ is the set of audio clap embeddings belonging to class c and $D_{A_i}^-$ is the set of audio clap embeddings belonging to sound classes other than c .

The final objective, $\mathcal{L}_{\text{Total}}$, is obtained by a weighted sum of all the three objective functions

$$\mathcal{L}_{\text{Total}} = \lambda \mathcal{L}_{\text{Ortho}} + \omega \mathcal{L}_{\text{Dev}} + \gamma \sum_c \mathcal{L}_{\text{INCE}}^c \quad (5)$$

s.t. $\lambda + \omega + \gamma = 1$

where λ , ω , and γ are investigated through experiments.

4. EXPERIMENTS AND RESULTS

We select 7 classes of "Ambient", "Animal", "Crowd", "Music", "Nature", "Speech", and "Wind" to evaluate the improvement in source separation performance achieved by our proposed framework. The selection of these classes was based on the AudioSet Ontology [24], the frequency of appearance in real-world videos, and inspired by the current commercially available options with similar functionality (e.g. Google Audio Magic Eraser and Samsung Audio Eraser). We applied the automated data engine on each dataset independently, as described in Section 3.1, with a target of 20K samples per sound class. The data engine generated more than 20K samples for the *speech* and *music* classes. In order to maintain a balanced class distribution, we randomly selected 20K samples from each class. In contrast, for classes such as *Crowd* and *Wind*, the combined number of

Target class	SDRi $\uparrow$			SI-SDRi $\uparrow$			CLAPScore $\uparrow$			CLAPScore _A $\uparrow$		
	Baseline	Refined	Optimized	Baseline	Refined	Optimized	Baseline	Refined	Optimized	Baseline	Refined	Optimized
Ambient	-1.623	4.116	4.301	-0.004	10.349	10.677	0.043	0.028	0.034	0.559	0.708	0.721
Animal	6.502	7.749	7.911	12.148	13.730	13.863	0.210	0.218	0.215	0.763	0.744	0.767
Crowd	2.507	4.060	4.026	8.449	10.596	10.561	0.145	0.120	0.123	0.794	0.830	0.843
Music	5.378	6.142	6.232	11.475	12.316	12.435	0.207	0.197	0.198	0.780	0.753	0.765
Nature	0.071	4.285	4.083	5.050	10.547	10.366	0.146	0.142	0.144	0.766	0.866	0.869
Speech	6.473	7.762	7.844	12.418	14.223	14.308	0.205	0.226	0.222	0.711	0.719	0.694
Wind	5.410	6.292	6.293	11.289	12.368	12.368	0.219	0.220	0.225	0.742	0.805	0.798
Average	2.753	5.545	5.610	7.386	11.798	11.904	0.148	0.143	0.145	0.707	0.763	0.770

Table 1. Performance comparison of the baseline, refined, and optimized control signals across target sound classes. Best results for each target class and metric are shown in bold. $\uparrow$ indicates that higher values are better.

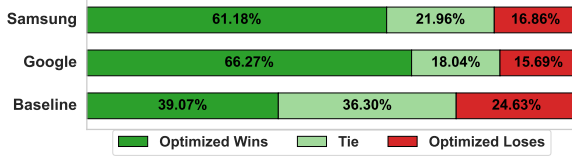


Fig. 2. Subjective preference evaluation of the separation quality for the model trained using optimized control signal.

audio samples across all the datasets did not reach 20K. We therefore supplemented the remaining samples using audio generative models and handcrafted techniques¹.

Since existing audio source separation models use a wide range of datasets for evaluation and there are no standard test set used consistently across the models, we selected a simple baseline architecture for our experiments. Specifically, we use AudioSep [9] model that introduces minimal architectural components beyond CLAP embeddings and FiLM layers as the control signal mechanism for single-channel source extraction. This choice allows us to evaluate the improvement in source separation through the proposed control signal optimization without additional architectural factor that could impact the comparison. We use Signal-to-Distortion Ratio improvement (SDRi) and the scale invariant version of it, SI-SDRi, as signal-level objective metrics to evaluate the performance of the source separation. We also use CLAPScore and CLAPScore_A as metrics to measure the semantic alignment of the extracted audio with respect to the caption (The sound of "Sound Class") and the audio samples of the same sound class in test set. We extend our evaluation to a subjective preference test among 17 participants to compare the sound separation quality of the models trained using our refined dataset and optimized control signal against the baseline model and the commercial models from Google and Samsung. We opted to do a listening test whereas the objective tests against the commercial options were not possible. This was due to the heavy processing applied on the audio signal when passing through the mobile apps which results in SDRi and SI-SDRi values of -7.668 and -34.914.

As shown in Table 3.2, both SDRi and SI-SDRi consistently improve from the baseline model to the refined model, with further improvement achieved by the optimized model. The improvements from the baseline to the refined model in-

dicating the benefits of the automated data engine for filtering and cleaning the training data, while further improvements from the refined to the optimized model highlight the effect of using the control signals obtained by the proposed optimization process. In contrast, CLAPScore shows minimal variation across three models. This may be attributed to the limited initial alignment between the CLAP embeddings and the target sound classes which further highlights the need for an optimization of CLAP-based control signals for source separation in scenarios with a limited set of target sound classes. CLAPScore_A, however, shows improvement from the baseline through the refined and optimized models, indicating the improved semantic similarity between the extracted audio and the target classes.

Figure 2 presents the results of the subjective evaluation, comparing the optimized model against the baseline model and the commercial options from Google and Samsung. Compared to the baseline model, the optimized model is preferred in 39.07% of the trials, has the loss rate of 24.63%, and 36.30% of the comparisons result in a tie. Our comparison with the commercial options also shows an overwhelming preference towards our model compared to both the models from Google and Samsung. Overall, the optimized model is preferred more frequently than it is disfavored in the subjective evaluations with both the baseline model and the commercial options. These results suggest that the proposed control-signal optimization provides a perceptually meaningful improvement in the separation quality.

5. CONCLUSION

Adapting LASS models to a defined set of general classes faces challenges such as training data procurement and lack of an effective control signal for source separation. We propose a framework consisting of an automated data engine and an optimization process for CLAP-based control signals to overcome these challenges. Through both subjective and objective tests we show that our framework is effective in adapting pretrained LASS models which outperforms both the baseline and commercial options. As future directions, we aim to explore other optimization methods for CLAP-based control signals and further refine the text prompt selection process.

6. REFERENCES

- [1] Ilya Kavalero, Scott Wisdom, Hakan Erdogan, Brian Patton, Kevin Wilson, Jonathan Le Roux, and John R Hershey, “Universal sound separation,” in *2019 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*. IEEE, 2019, pp. 175–179.
- [2] Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang, “Clap learning audio concepts from natural language supervision,” in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [3] Hao Ma, Zhiyuan Peng, Xu Li, Mingjie Shao, Xixin Wu, and Ju Liu, “Clapsep: Leveraging contrastive pre-trained model for multi-modal query-conditioned target sound extraction,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 4945–4960, 2024.
- [4] Tanvir Mahmud, Saeed Amizadeh, Kazuhito Koishida, and Diana Marculescu, “Weakly-supervised audio separation via bi-modal semantic similarity,” *International Conference on Learning Representations*, vol. 2024, pp. 29115–29147, 2024.
- [5] Hae Chun Chung and Jae Hoon Jung, “Language-queried audio source separation enhanced by expanded language-audio contrastive loss,” *DCASE2024 Challenge, Tech. Rep.*, 2024.
- [6] Hao Ma, Zhiyuan Peng, Xu Li, Yukai Li, Mingjie Shao, Qiuqiang Kong, and Ju Liu, “Language-queried target sound extraction without parallel training data,” in *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pp. 1–5.
- [7] Jianyuan Feng, Guangzheng Li, and Yangfei Xu, “Hybrid-sep: Language-queried audio source separation via pre-trained model fusion and adversarial consistent training,” in *2025 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*. IEEE, 2025, pp. 1–5.
- [8] Jeong Yeol Yang, Hyun Joon Park, Soyoung Lee, and Sung Won Han, “Clapmamba: Clap-guided mamba for language-queried target sound extraction,” *IEEE Access*, 2026.
- [9] Xubo Liu, Qiuqiang Kong, Yan Zhao, Haohe Liu, Yi Yuan, Yuzhuo Liu, Rui Xia, Yuxuan Wang, Mark D Plumbley, and Wenwu Wang, “Separate anything you describe,” *IEEE Transactions on Audio, Speech and Language Processing*, vol. 33, pp. 458–471, 2024.
- [10] Kohei Saijo, Janek Ebbers, François G Germain, Sameer Khurana, Gordon Wiehern, and Jonathan Le Roux, “Leveraging audio-only data for text-queried target sound extraction,” in *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pp. 1–5.
- [11] Zihan Zhang, Xize Cheng, Zhennan Jiang, Dongjie Fu, Jingyuan Chen, Zhou Zhao, and Tao Jin, “Mars-sep: Multimodal-aligned reinforced sound separation,” in *International Conference on Learning Representations*, 2026, vol. 2026, pp. 83453–83477.
- [12] Alexandre Défossez, Nicolas Usunier, Léon Bottou, and Francis Bach, “Music source separation in the waveform domain,” *arXiv preprint arXiv:1911.13254*, 2019.
- [13] Simon Rouard, Francisco Massa, and Alexandre Défossez, “Hybrid transformers for music source separation,” in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [14] Yi Luo and Jianwei Yu, “Music source separation with band-split rnn,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 1893–1901, 2023.
- [15] Minseok Kim, Woosung Choi, Jaehwa Chung, Daewon Lee, and Soonyoung Jung, “Kuielab-mdx-net: A two-stream neural network for music demixing,” *arXiv preprint arXiv:2111.12203*, 2021.
- [16] Yi Luo and Nima Mesgarani, “Conv-tasnet: Surpassing ideal time-frequency magnitude masking for speech separation,” *IEEE/ACM transactions on audio, speech, and language processing*, vol. 27, no. 8, pp. 1256–1266, 2019.
- [17] Ke Chen, Jiaqi Su, and Zeyu Jin, “Mdx-gan: Enhancing perceptual quality in multi-class source separation via adversarial training,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 741–745.
- [18] Yinkai Zhang, Dingbang Zhang, Tao Wang, Diana Rakhimova, Kai Wang, and Hao Huang, “Domain partitioning meets parameter-efficient fine-tuning: A novel method for improved language-queried audio source separation,” in *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2026, pp. 15637–15641.
- [19] Zafar Rafii, Antoine Liutkus, Fabian-Robert Stöter, Stylianos Ioannis Mimilakis, and Rachel Bittner, “Musdb18-hq - an uncompressed version of musdb18,” Aug. 2019.
- [20] Joris Cosentino, Manuel Pariente, Samuele Cornell, Antoine Deleforge, and Emmanuel Vincent, “Librimix: An open-source dataset for generalizable speech separation,” *arXiv preprint arXiv:2005.11262*, 2020.
- [21] Xubo Liu, Haohe Liu, Qiuqiang Kong, Xinhao Mei, Jinzheng Zhao, Qiushi Huang, Mark D. Plumbley, and Wenwu Wang, “Separate What You Describe: Language-Queried Audio Source Separation,” in *Interspeech 2022*, 2022, pp. 1801–1805.
- [22] Kevin Kilgour, Beat Gfeller, Qingqing Huang, Aren Jansen, Scott Wisdom, and Marco Tagliasacchi, “Text-Driven Separation of Arbitrary Sounds,” in *Interspeech 2022*, 2022, pp. 5403–5407.
- [23] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville, “Film: Visual reasoning with a general conditioning layer,” in *Proceedings of the AAAI conference on artificial intelligence*, 2018, vol. 32.
- [24] Jort F Gemmeke, Daniel PW Ellis, Dylan Freedman, Aren Jansen, Wade Lawrence, R Channing Moore, Manoj Plakal, and Marvin Ritter, “Audio set: An ontology and human-labeled dataset for audio events,” in *2017 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2017, pp. 776–780.
- [25] Chris Dongjoo Kim, Byeongchang Kim, Hyunmin Lee, and Gunhee Kim, “Audiocaps: Generating captions for audios in the wild,” in *NAACL-HLT*, 2019.
- [26] Honglie Chen, Weidi Xie, Andrea Vedaldi, and Andrew Zisserman, “Vggsound: A large-scale audio-visual dataset,” in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 721–725.
- [27] Eduardo Fonseca, Jordi Pons Puig, Xavier Favory, Frederic Font Corbera, Dmitry Bogdanov, Andres Ferraro, Sergio Oramas, Alastair Porter, and Xavier Serra, “Freesound datasets: a platform for the creation of open audio datasets,” in *Hu X, Cunningham SJ, Turnbull D, Duan Z, editors. Proceedings of the 18th ISMIR Conference; 2017 oct 23-27; Suzhou, China.[Canada]: International Society for Music Information Retrieval; 2017. p. 486-93. International Society for Music Information Retrieval (ISMIR)*, 2017.
- [28] British Broadcasting Corporation, “Bbc sound effects,” <https://bbcrewind.co.uk>, 2018, Accessed: 2026-08-17.
- [29] Roland S. Zimmermann, Evgenia Rusak, Wieland Brendel, Attila Juhas, Patrik Reizinger, and Oliver Bringmann, “InfoNCE: Identifying the gap between theory and practice,” in *High-dimensional Learning Dynamics 2024: The Emergence of Structure and Reasoning*, 2024.